Alexander Lavrenenko he/him/his

(228) 334-9939	alavrenenko.com	lavrenenko.2003@gmail.com	linkedin.com/in/alexander-lavrenenko

EDUCATION

University of Massachusetts Amherst | College of Information and Computer Sciences	Exp. Graduation May 2024
B.S. in Computer Science and Mathematics (double major)	GPA: 4.0

Coursework: Software Engineering, Data Structures (Java), Algorithms, Programming Methodology (JavaScript), Discrete Mathematics, Multivariable Calculus, Linear Algebra

SKILLS

Programming Languages: Java, Kotlin, Python, JavaScript, C++, MATLAB, GLSL, CSS

Tools: TDD, SOLID, OOP, SQL, GitHub, APIs, Bazel, Slack, Linux Command Prompt, Multithreading, UI/UX, File Management, Firebase, TensorFlow, Pandas, SK Learn, Retrofit, Ktor

IDE's: Android Studio, Visual Studio Code, IntelliJ Idea, PyCharm, Jupyter Notebook, NetBeans, Atom

Awards: Faculty Honors in Fall 2021 and Spring 2022, multiple prize-winner of Russian National Physics Olympiad, Physics and Mathematics Olympiad "Phystech", Mathematics Olympiad "Unity Formula"

Clubs: UMass ACM Machine Learning Club, UMass Undergraduate Math Club

WORK EXPERIENCE

University of Massachusetts Amherst	Jul – Aug 2022
Manager - UMass Summer Conference Services	Amherst, MA

- Handled responsibility for the apartment's preparation, resolution of conflicts between clients and supervision of access cards. Handled over 60+ lockouts and other conflicts, received positive feedback from supervisors and guests from over 12+ conferences.

MoodMe	Feb – Apr 2022
Android Developer Intern	Boston, MA

- Founded project to develop a module for android to remove the user's background. Enhanced efficiency by 60% building with Bazel on C++ in Visual Studio and using open-source Google Mediapipe library.

Russian Academy of Sciences	Jun 2020 – Sep 2020
Software Engineer Intern	Moscow, Russia

- Automated tedious manual task by developing algorithm to solve Gaussian peaks obtained with Roman spectroscopy made with MATLAB and Simulink. Improved it by saving results into separate file for later processing.

TAU Tracker Company	Nov 2019 – Oct 2020
Software Engineer Intern	Moscow, Russia

- Enhanced magnetic coil simulation algorithms written on MATLAB and improved their speed by 25% and accuracy by over 30%. Received positive feedback from founders of the startup.

PROJECTS

github.com/Scrip00

3D Browser Engine — github.com/Scrip00/3d_browser_engine

- Developed 3D browser engine using pure JavaScript, WebGL library and GLSL shader. Coded efficient raytracing algorithms and implemented multiple materials including glass, metal and matte objects or even 3D fractals.

UMass Maps application — github.com/Scrip00/umassmaps

- Made advanced maps application with Kotlin in Android studio using SOLID and TDD principles and MVVM pattern.
- Technologies: Room DB, Retrofit HTTP Client, External APIs, Firebase Firestore, Ktor asynchronous server.

SkinSafe application – 48hr hackathon team project — github.com/Scrip00/SkinSafe

- Handled responsibility for developing Android application in Java which uses AI to detect and classify among most popular skin lesions preventing people from deaths due to unrecognized in time diseases and published in Play Store.
- Technologies: Java, Python, TensorFlow, Android Studio, Jupyter Notebook, Room DB, Material Design, CNN.

Selfie Segmentation Library — github.com/Scrip00/Selfie-segmentation-library

- Built open-source selfie segmentation Android library with Bazel based on Google Mediapipe library using C++. The library includes a lot of prebuilt features like image and video background or static and dynamic solid color.

Vaccinations Tracking App — github.com/Scrip00/vaccinations_tracking_app

- Developed desktop application to help the school secretary monitor re-vaccination and check-up dates and automatically notify staff via email using Java. Freed the manager from manual work and improved the efficiency of notifications. Got positive feedback from end-users and school managers.

AI lab – *the collection of projects* — github.com/Scrip00/AI-lab

- Mastered Machine Learning techniques with projects including watermark removal via Convolutional Autoencoder and photo to Picasso stylizer via Cycle-Consistent Adversarial Networks.
- Technologies: CNN, Natural Language Processing, Pix2Pix, CycleGAN, Convolutional Autoencoders, Image Segmentation, Data Augmentation, Data Preprocessing, Python, TensorFlow, Keras, Pandas, Jupyter Notebook.

Magician AI — github.com/Scrip00/Magician-AI

- Developed Machine Learning model to guess user’s card by highlighting deviations in his emotions and tone of speech. Made a desktop app with Java via NetBeans IDE to effectively collect training data.

Numble Game — github.com/Scrip00/Numble

- Developed mobile game on Java where user should guess randomly generated equation. Came up with unique equation generator algorithm involving even non-trivial mathematical operations and published in Google Play Store.

LEADERSHIP EXPERIENCE

Computer Science Club	Founder and President	Fall 2019 – Spring 2021

- Included 5 members, lead weekly meetings (1-2 hours) for members to share ideas, discuss trends, and work on research. Helped 5 members to enhance their coding skills and implement them in their personal projects.

Skolkovo International Gymnasium	Mentor	Fall 2019 – Spring 2021

- Peer mentoring: G7 students in Math; G9 students in Physics and Math; conducted lessons for kindergarten students, personally mentored 6+ students. Received positive feedback from school teachers about mentees academic success.

Skolkovo School Volunteering Club	Member and Teaching Assistant	Fall 2019 – Spring 2021

- Helped school organizers to host a Charity Fair; dedicated time to help children on the Autism spectrum by working as a special education teaching assistant. Personally trained 4+ children with ASD and improved their socialization skills.